



Exposure to AI Overviews and News Consumption

ITT-style difference-in-differences pre-analysis plan

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Motivation

- Google's AI Overviews (AIO) launched in the US on **May 14, 2024**, rendering answers and summaries inline above blue links.
- Users' **information-seeking behaviors** are shifting in response to AI-powered search tools and chatbots.
- Displacement effect : AI summary halved click-through rate (Pew Research Center, 2025)
- Political communication research shows search-mediated browsing to be a primary gateway to incidental news exposure
- Yet no existing study estimates whether repeated AIO exposure cumulatively shifts users' news-reading baseline across time.

Research Question

RQ1

Does AIO exposure shift user-level news consumption?

RQ2

Which AIO text features drive click-through?*(Text-as-treatment using sIBP / Fong-Grimmer)

*This poster is only presenting analysis plan and simulation for RQ1

Data & Setup

Intended Dataset

- National Internet Observatory browsing data (panel)

Simulation Data

- Google search data Jan 2024 – Jan 2025
- Time-on-page data: Jan 2024 – Jan 2025
- News domain filtering:
News domain quality rating (Lin et al. 2013) filter by Media Bias/Fact Check (MBFC) rating > 0.5

Setup

Balanced panel: N = 402 users × 35 weeks.

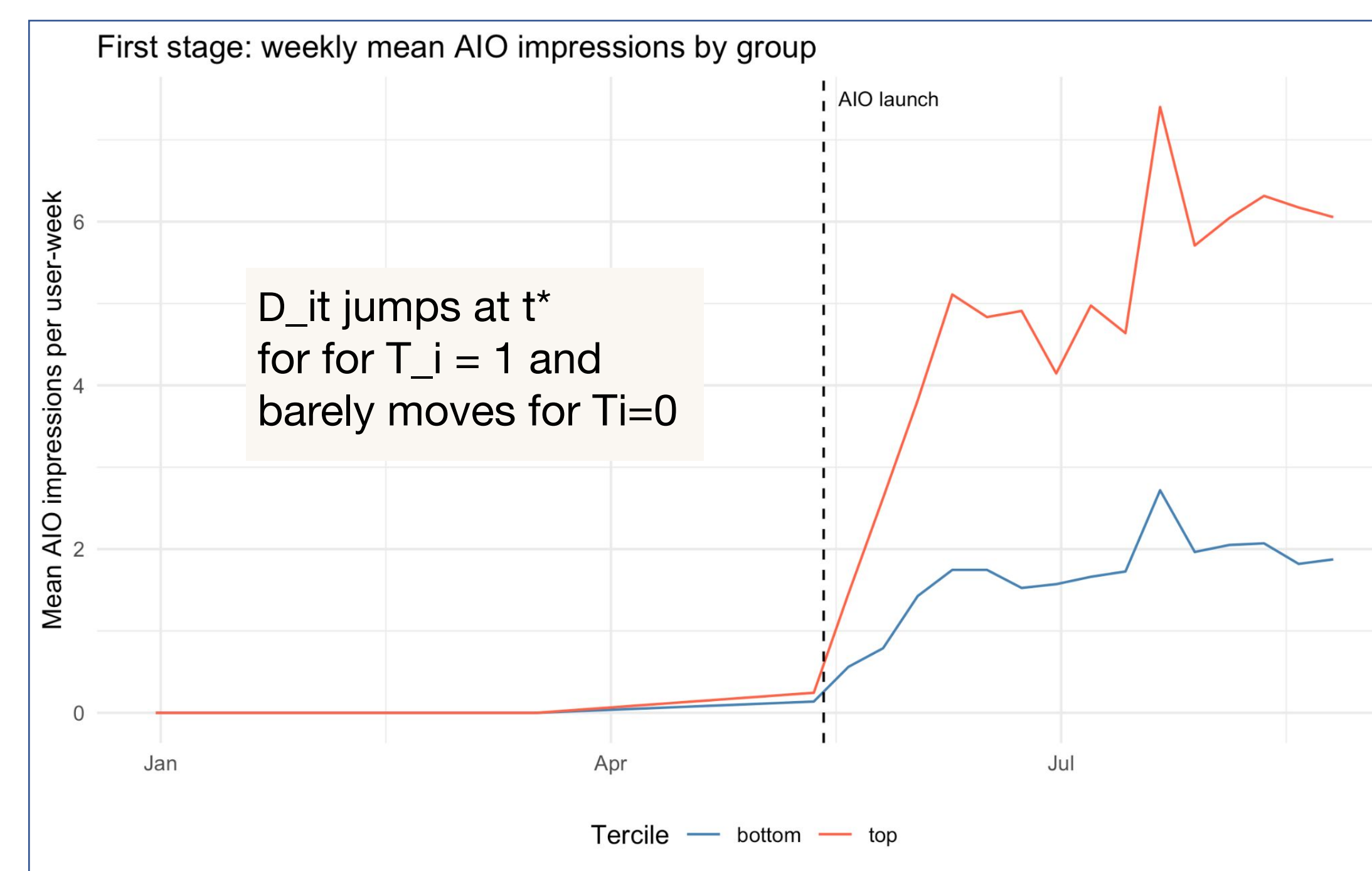
Window: **Jan 1 – Aug 31, 2024**, with limited-testing weeks (Mar 22 – May 13) dropped.

i, t	user and week indices
t^*	treatment time: week containing May 14, 2024
Y_{it}	outcome: $\log(1 + \text{news-domain seconds in week } t)$
s_i^{pre}	pre-period informational-query share (Jan 1 – Mar 21 only)
$T_i \in \{0,1\}$	treated = top tercile of s_i^{pre} ; control = bottom; bottom tercile of s_i^{pre} ; middle dropped
Post_t	$= 1\{t \geq t^*\}$
D_{it}	AIO impressions for user i in week t (<i>exposure dose, not the treatment itself</i>)
α_i, λ_t	user and week fixed effects
$X_{i,t-1}$	lagged time-varying covariates (log total online time) — secondary spec, potentially endogenous

Design & Identification

First stage: ITT-style exposure shock

Design logic: The AIO rollout is a *uniform* shock — no user is randomized into AIO at t^*
 T_i : indicator of post-launch AIO exposure intensity **based on pre-period informational query share**



$$\Delta D = E[D_{it} | T=1, t \geq t^*] - E[D_{it} | T=0, t \geq t^*] \approx 5.0 - 1.8 = 3.2 \text{ impressions / user-week } (\approx 2.8\times \text{ ratio})$$

2*2 DiD Framework

Two-way fixed-effects regression:

$$Y_{it} = \alpha_i + \lambda_t + \tau \cdot (T_i \times \text{Post}_t) + X'_{i,t-1}\beta + \varepsilon_{it}$$

SEs clustered at user level (Nclusters = 402); cluster-bootstrap for SDID.

Estimand & Interpretation

Estimand: τ is the *ITT-style reduced-form ATT* under parallel trends:

$$\tau = E[Y_{it}(1) - Y_{it}(0) | T_i = 1, t \geq t^*]$$

what τ is : the average post-launch change in log news-time among high-informational-share users, *minus* the counterfactual change they would have experienced had AIO never launched.
Where the bottom tercile's change is used as the counterfactual.

Threats & Robustness

Confounds:

- 2024 election cycle, campaign peak
- Biden→Harris transition (Jul 21)
- Control still gets some AIO dose

Selection Bias:

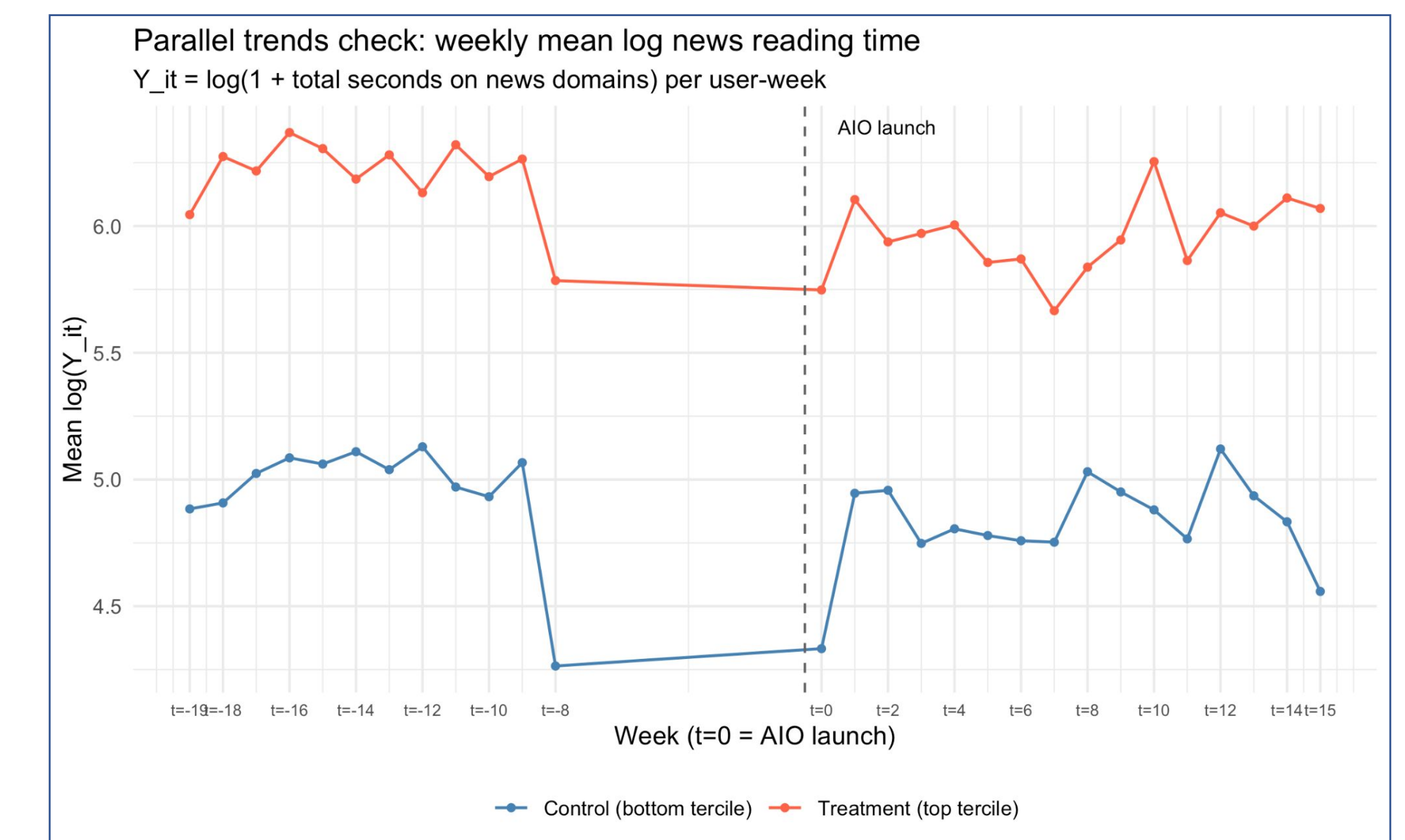
Treatment and group systematically different in baseline information seeking and news reading behavior

Planned Checks:

placebo outcome (non-news domain) ; pre-period placebo (fake t^*)

Simulation Results

Parallel-trends test



Pre-period slopes $\approx$ parallel; levels differ as expected

2x2 DiD estimates

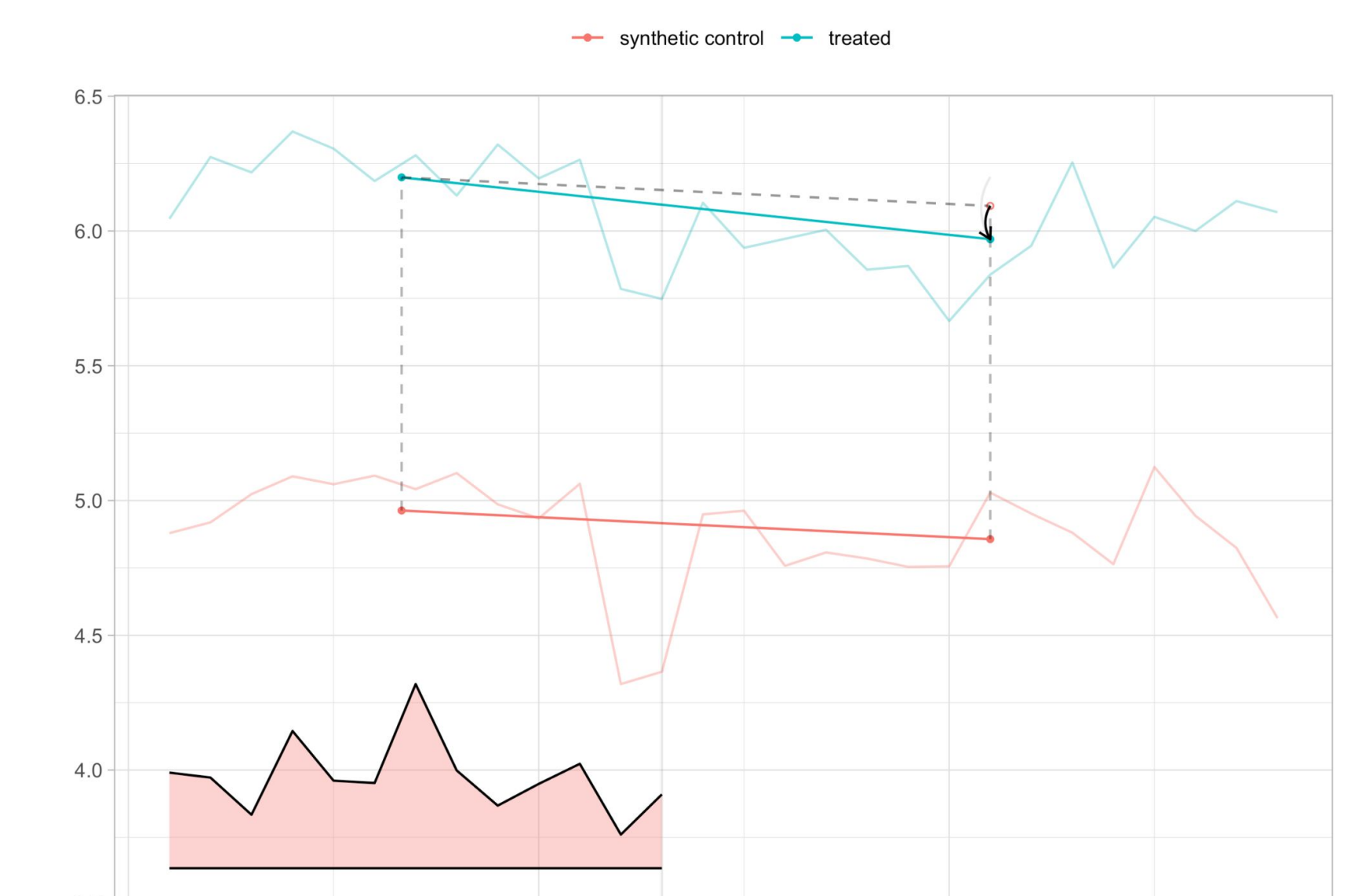
Specification	τ_{hat}	SE (cluster)
Primary (no covariates)	-0.140**	(0.050)
+ lagged log online time	-0.156**	(0.051)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. SEs clustered at user level (N = 402).

$\approx 13\text{--}14\%$ reduction in news-domain time for high-AIO-exposure-propensity users (reduced-form ITT)

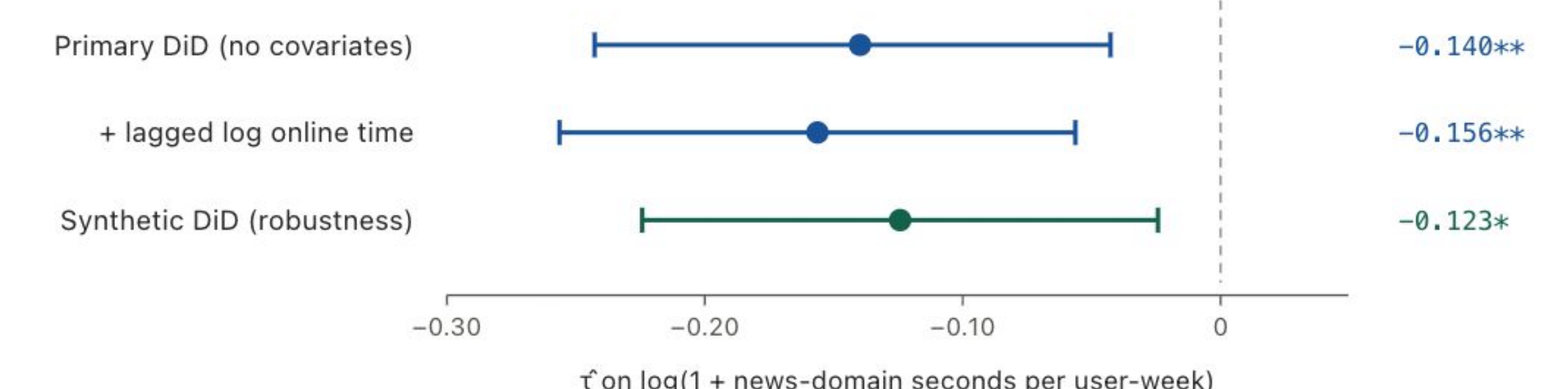
Synthetic DiD (robustness)

Estimate unit weights $\hat{w}_i$ (so weighted control's pre-period news-reading trajectory best parallels the treated) and time weights $\hat{\lambda}_t$ (up-weight post-like pre-weeks)



SDID: synthetic control matches treated pre-trajectory; effect ≈ -0.12 .

Treatment effect estimates with 95% confidence intervals



References

- Chapekis, Athena, and Anna Lieb. 2025. "Google Users Are Less Likely to Click on Links When an AI Summary Appears in the Results." Pew Research Center, July 22, 2025. <https://www.pewresearch.org/short-reads/2025/07/22/google-users-are-less-likely-to-click-on-links-when-an-ai-summary-appears-in-the-results/>
- Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. 2021. "Synthetic Difference-in-Differences." American Economic Review 111 (12): 4088–4118. DOI: 10.1257/aer.20190159